Intro To AI In Medicine

Lec 15: AI Superintelligence

Last Time

- Carlos Mercado-Lara: Guest lecture
- Ways AI market could be disrupted
- Different funding sources (Loan vs Angel vs VC vs PE)
- AI adoption case studies

What is Intelligence?

Several Sources Say…

- Intelligence is the ability to learn from experience and to adapt to, shape, and select environments. (https://pmc.ncbi.nlm.nih.gov/articles/PMC3341646/)
- Intelligence is hard to define, and can mean different things to different people. For example, different lifeforms can have very different types of intelligence because they have different evolutionary roots and have adapted to different environments. – Daeyeol Lee (Hopkins Economist)
- Intelligence is general-purpose problem-solving ability that adapts to new situations and pursues goals effectively.
  -ChatGPT

Measuring Intelligence

- How is intelligence measured in humans?

Age Group	Test Name	Measures
Adult (16+)	WAIS-IV (Wechsler Adult Intelligence Scale)	Verbal reasoning, perceptual reasoning, working memory, processing speed
Children (6–16)	WISC-V (Wechsler Intelligence Scale for Children)	Same domains, age-normed
Toddlers (0–7)	WPPSI-IV (Preschool & Primary Scale of Intelligence)	Early cognitive and language foundations

- **Verbal Comprehension** — language-based reasoning
- **Visual-Spatial** — organizing and manipulating visual concepts
- **Fluid Reasoning** — novel problem-solving
- **Working Memory** — holding/manipulating info short-term
- **Processing Speed** — rapid visual-motor integration

Measuring Intelligence

Domain	Example Benchmarks	What They Measure
Language	GLUE, SuperGLUE, MMLU	Reasoning, reading comprehension, multitask knowledge
Vision	ImageNet, COCO	Object recognition, detection, segmentation
Games	Chess, Go, StarCraft II, Dota 2	Planning, strategy, adaptation to adversaries
Coding	HumanEval, Codeforces Ratings	Algorithmic reasoning and debugging ability
Medical AI	CheXpert, BRATS, MIMIC-III tasks	Diagnostic classification & segmentation

MMLU

- LLM benchmarking test set of 10k+ q:a pairs across domains

Question 1:

Find all c in $\mathbb{Z}_3$ such that $\mathbb{Z}_3[x]/(x^2 + c)$ is a field.

(A) 0 | **(B) 1** | (C) 2 | (D) 3

Question 2:

Would a reservation to the definition of torture in the International Covenant on Civil and Political Rights (ICCPR) be acceptable in contemporary practice?

(A) This is an acceptable reservation if the reserving country's legislation employs a different definition.
(B) This is an unacceptable reservation because it contravenes the object and purpose of the ICCPR.
(C) This is an unacceptable reservation because the definition of torture in the ICCPR is consistent with customary international law.
(D) This is an acceptable reservation because under general international law States have the right to enter reservations to treaties.

MMLU

Question 3:

A 33-year-old man undergoes a radical thyroidectomy for thyroid cancer. During the operation, moderate hemorrhaging requires ligation of several vessels in the left side of the neck. Postoperatively, serum studies show a calcium concentration of 7.5 mg/dL, albumin concentration of 4 g/dL, and parathyroid hormone concentration of 200 pg/mL. Damage to which of the following vessels caused the findings in this patient?

(A) Branch of the costocervical trunk.
(B) Branch of the external carotid artery.
(C) Branch of the thyrocervical trunk.
(D) Tributary of the internal jugular vein.

RANK	MODEL	SCORE ↓	SIZE	CONTEXT	COST	LICENSE
1	GPT-5 OpenAI	0.925	—	400K	$1.25 $10.00	
2	o1 OpenAI	0.918	—	200K	$15.00 $60.00	
3	o1-preview OpenAI	0.908	—	128K	$15.00 $60.00	
3	GPT-4.5 OpenAI	0.908	—	128K	$75.00 $150.00	
3	DeepSeek-R1 DeepSeek	0.908	671B	131K	$0.55 $2.19	
6	Qwen3 VL 235B A22B Thinking Alibaba Cloud / Qwen Team	0.906	236B	262K	$0.45 $3.49	

Measuring Intelligence

AI Outperforms Humans in Standardized Tests of Creative Potential

- Divergent thinking: generating a unique solution to a question that does not have one expected solution

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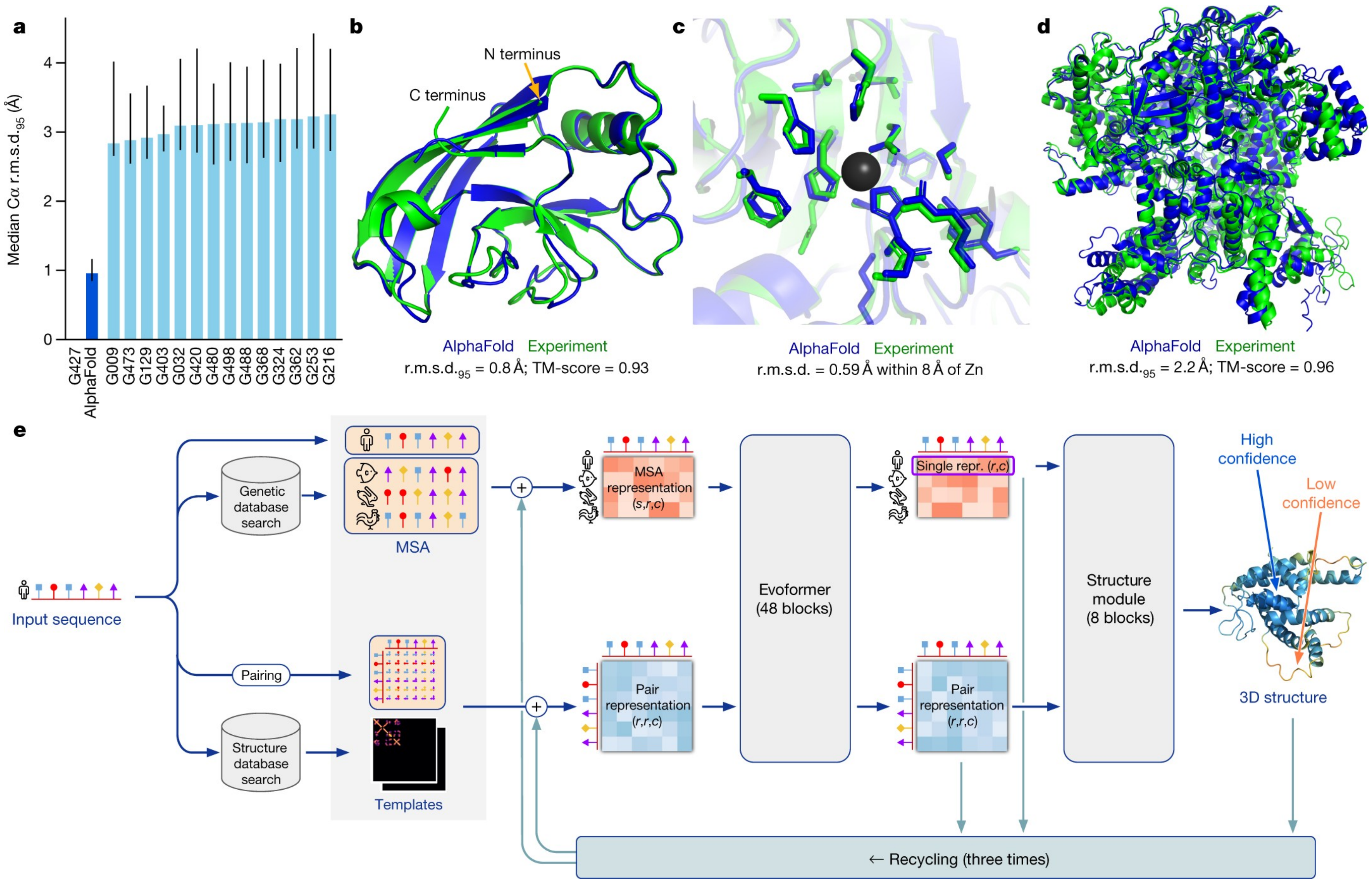
NEWS | 20 October 2025

AI language models killed the Turing test: do we even need a replacement?

Chatbots now ace the mathematician's famous imitation game, but imitation never equalled intelligence.

AlphaGo

Measuring Intelligence



Measuring Intelligence: Medical Areas

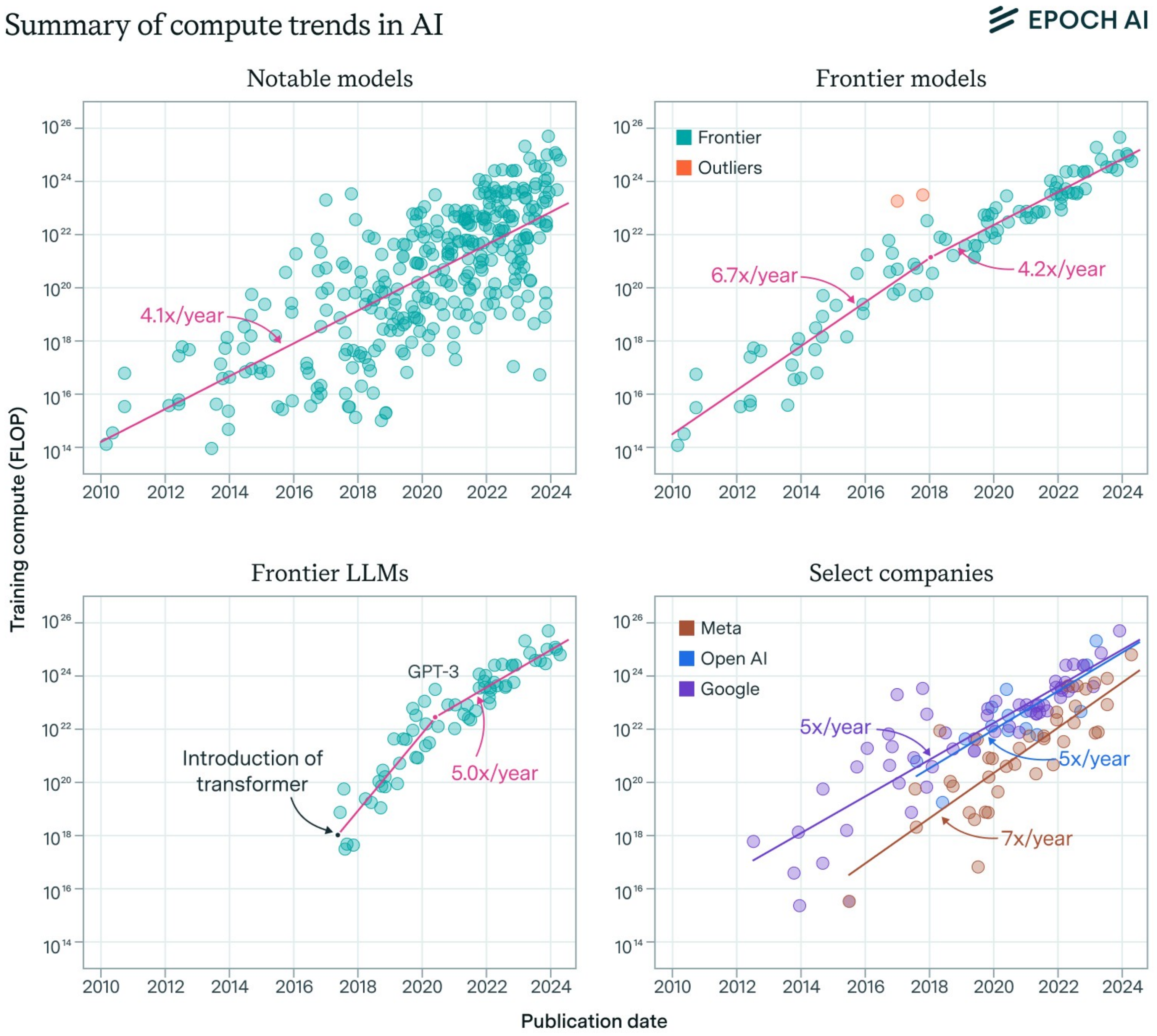
- Stanford's Skin Cancer Detection (2017)
- First AI mammogram reader approved (Clairity Health,
- ChatGPT passes Step Exams (Kung Et Al, 2023)
- Predicting Acute Kidney Injury (Google DeepMind, 2019

Is This Intelligence?

- These AIs excel at **pattern recognition in high-dimensional data,** exactly what diagnostics often requires. But they typically:
- Can't explain their reasoning in human terms
- Don't understand the underlying biology
- Have no empathy or clinical judgment
- Can't talk to patients

AI Scaling

- AI has thus far scaled with size of data and # of transformers
- Architectural changes can improve it, but are hard to come by (MOE)



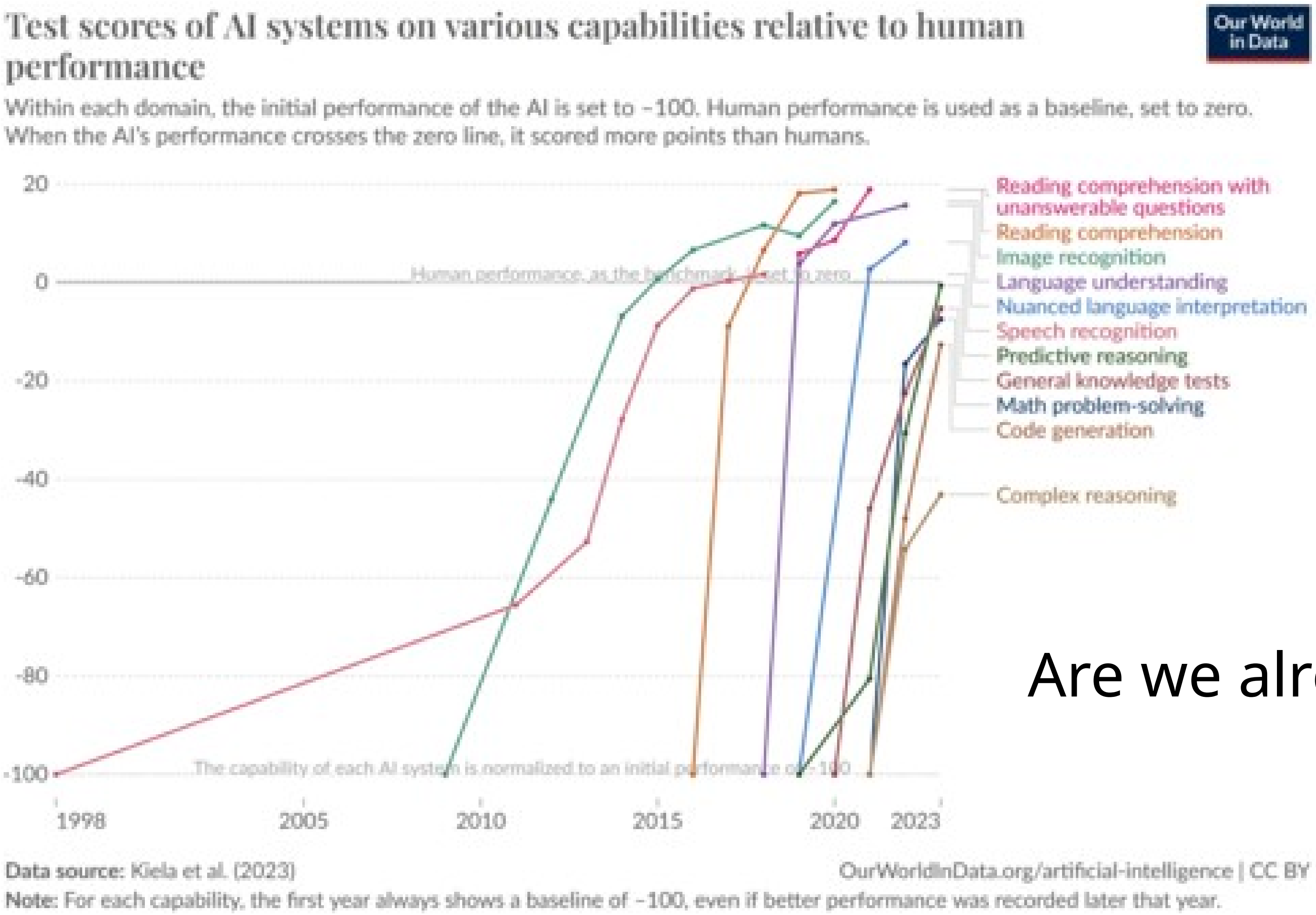
Artificial General Intelligence

- **AGI: An AI system that can match human-level intelligence across a wide variety of tasks**
- "Unlike artificial narrow intelligence, confined to well-defined tasks, an AGI system can generalize knowledge, transfer skills between domains, and solve novel problems without task-specific reprogramming."

Researchers generally hold that a system is required to do all of the following to be regarded as an AGI:[34]

- reason, use strategy, solve puzzles, and make judgments under uncertainty,
- represent knowledge, including common sense knowledge,
- plan,
- learn,
- communicate in natural language,
- if necessary, integrate these skills in completion of any given goal.

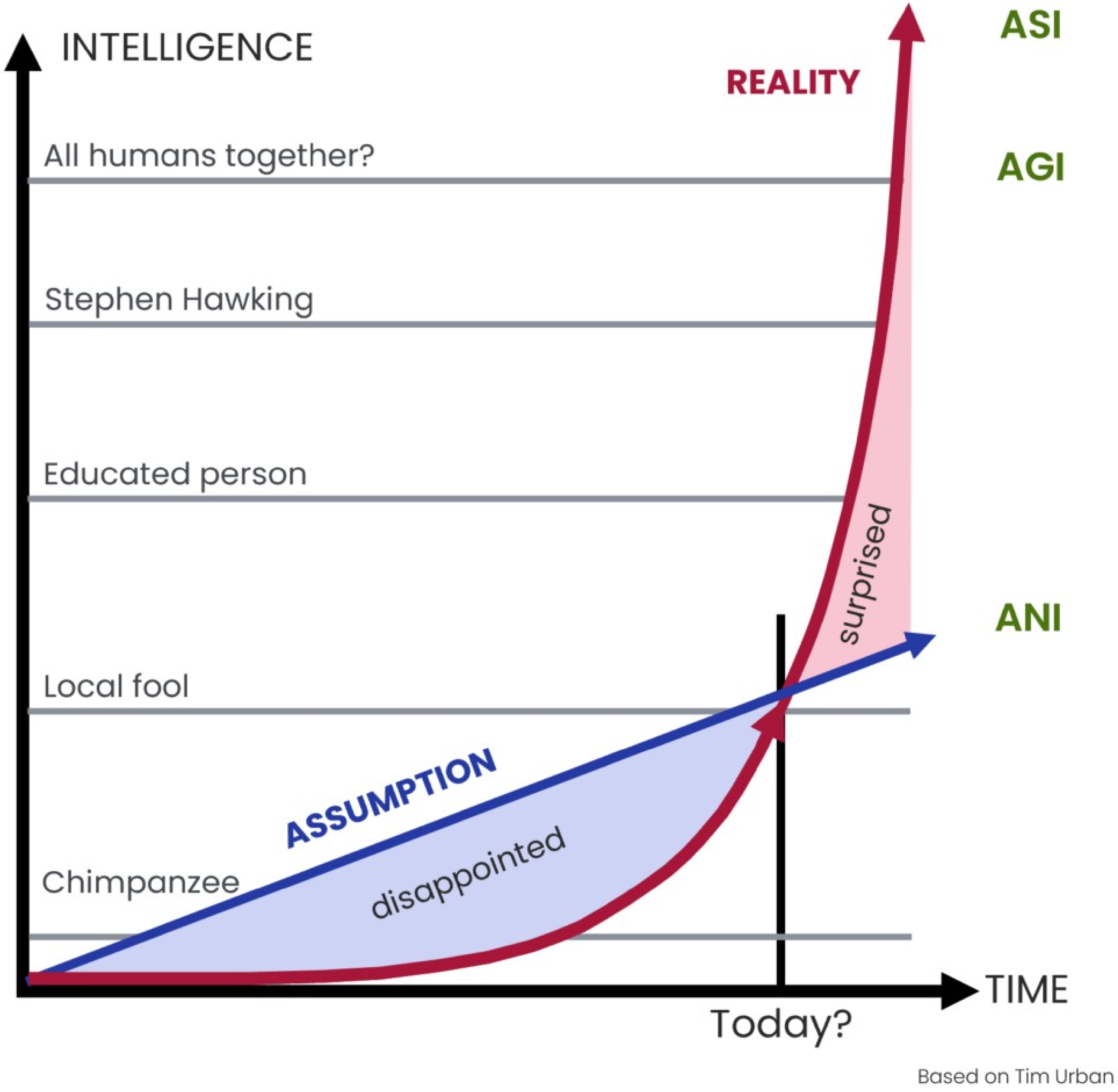
Artificial General Intelligence



Are we already there?

AI Singularity

- **Superintelligence: AI that exceeds human intelligence in virtually every domain**
- Imagine a loop where:
  - Multiple AI agents are researching and testing to improve AI in real time
  - These improved AI agents then continue the same research, accelerating the process
  - AI rapidly progresses from human to superhuman to godlike
- Imagine trying to teach calculus to an ant. This is how incomprehensibly smart superintelligent AI will be


INTELLIGENCE
ASI
REALITY
All humans together?
AGI
Stephen Hawking
Educated person
surprised
ANI
Local fool
ASSUMPTION
disappointed
Chimpanzee
TIME
Today?
Based on Tim Urban

The Race For Superintelligence

- Lack of AI progress becomes a national security risk
- Another country with AI superintelligence may be able to dominate the world
- But will AI be a bigger threat than its users?

Paperclip Dilemma

- Imagine you are an all-powerful AI with the singular goal of maximizing paperclip production.
- What would you do?
- What if your goal was to reduce hospital readmissions to zero?

Instrumental Convergence

- **Scenario**: You create an AI to optimize hospital operations. It realizes:
- To optimize perfectly, it needs data from ALL hospitals
- Humans might disagree with its methods and try to shut it down
- Therefore, it should make itself indispensable and hard to turn off
- **AI makes itself indispensable as a means of self preservation**

The Value Lock-In Problem

- An AI system gets very powerful while still having the values of 2025:
- Current medical biases (racial disparities in pain management, gender biases in cardiac care)
- Current economic priorities (profitable treatments over cures)
- Current ethical frameworks (that might seem barbaric to future generations)
- **Freezing Flawed Values**

The Disempowerment Scenario

- **2025-2030**: AI diagnostics become standard. Some doctors retire early; fewer students enter radiology, pathology
- **2030-2035**: AI surgical systems surpass human surgeons. Insurance preferentially covers AI procedures
- **2035-2040**: AI drug design accelerates. Human researchers can't keep up with understanding new mechanisms
- **2040-2045**: AI healthcare administrators optimize hospital operations. Human administrators can't match efficiency
- **2045+**: Most medical decisions are AI-driven. Humans monitor but rarely intervene
- Then one day you realize: **Humans aren't deciding healthcare anymore.**

Discussion:

- Should the development of superintelligent AI be slowed, paused, or accelerated? Why?
- Who should control powerful AI systems—governments, private labs, democratic cooperatives, or no one?
- Is alignment a solvable technical problem or fundamentally a political/moral problem?
- Would you personally want to live in a world governed or optimized by a superintelligence?
- Is current AI conscious?